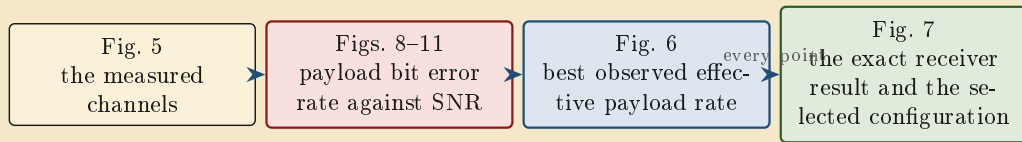


Reading the Red Result Figures

The measured channels, the payload bit error rate curves, the best observed rate, and the two maps that explain every point



The measured channels

the red captures of the Underwater Acoustic Channel Repository

one deployment
transmitter
ceiver 4.6 m,
rier 25 kHz,

a horizontal
array, 0.6 m spacing

capture	hyd.	Doppler	variability	0.9-corr. time	30 dB del
red 1	1-3	+2.30 to +2.32 Hz	0.54-0.62 Hz	156.2-187.5 ms	15.68-32.76 m
red 2	1-3	+2.14 to +2.15 Hz	0.64-0.89 Hz	125.0-135.4 ms	14.95-38.85 m
red 3	1-3	+1.46 to +1.47 Hz	0.43-0.53 Hz	166.7-187.5 ms	10.10-38.02 m
red 4	1-3	-0.24 to -0.23 Hz	0.75-1.05 Hz	145.8-166.7 ms	7.66-9.79 m

the paper's Table II now gives all twelve capture-hydrophone rows; the delay column is the span within 30 dB

the mean residual Doppler falls monotonically from +2.3 Hz to -0.2 Hz across the four captures; at the 25 kHz carrier, +2.3 Hz is a steady drift near 0.14 m/s, and the last capture is nearly stationary

Four captures share this deployment record and differ only in their codenames; their measured statistics separate them. Red 2 is the fastest path — watch its row when the bit error rate figures arrive.

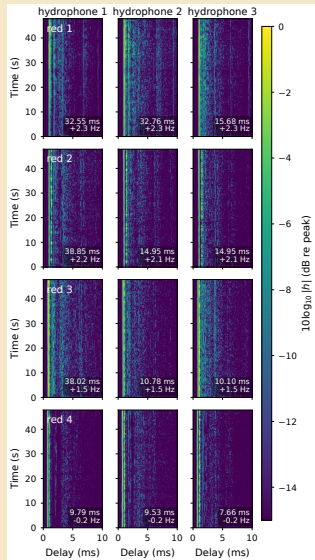


Fig. 5 of juna_joint_ieee.tex

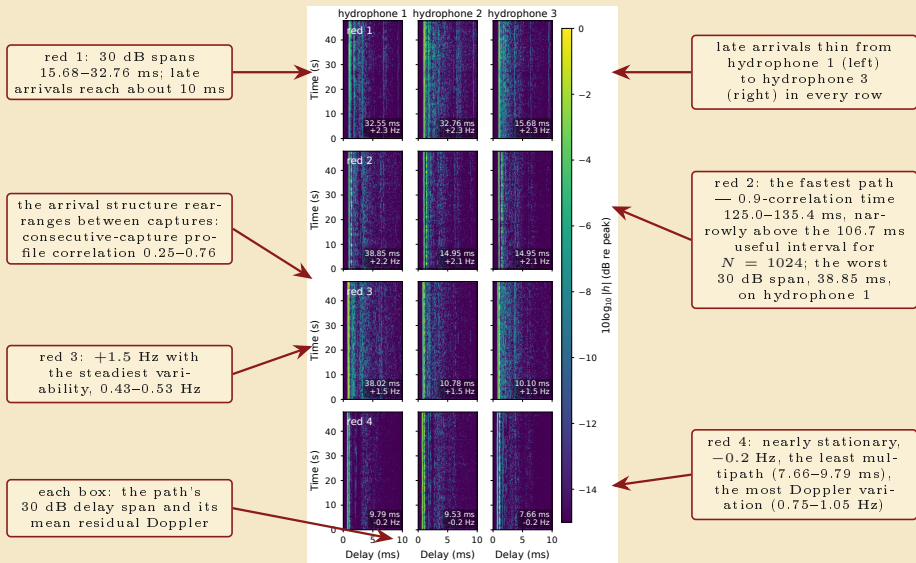
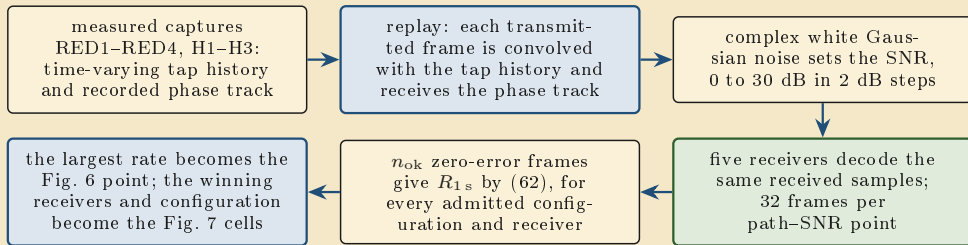


Fig. 5 of juna_joint_ieee.tex

From capture to point



Five receivers decode the same received samples at every point, so each comparison holds the frames, the channel, and the noise fixed. Eighteen admitted configurations compete at every path-SNR point.

The grid

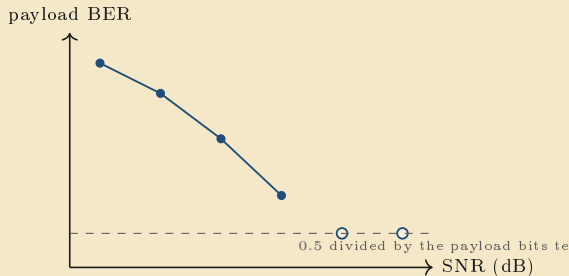
	H1	H2	H3
RED1			
RED2			
RED3			
RED4			

rows give RED1–RED4:
the four captures

columns give H1–H3:
the three hydrophones

Twelve panels, one per capture–hydrophone path. Each panel is read the same way; the marked panel is RED2 on H2.

The payload bit error rate figures



twelve capture-hydrophone panels; the vertical axis is payload bit error rate against SNR

at points with no observed bit error, hollow markers sit at 0.5 divided by the payload bits tested

this floor is a measurement limit, not a measured bit error rate

The panel here is schematic; the printed figures carry the data. The JUNA-C, z and JUNA-Direct-C, z receivers return a joint candidate only when it passes the 16-bit CRC; otherwise each returns its fallback output.

One selected configuration per N

Figs. 8–11 show one comparison for each displayed N : 1024, 512, 1536, and 2048

the configuration for each N is selected by its mean configured-one-second effective payload rate over the twelve paths and sixteen sampled SNR values

at each path–SNR point, the selection first takes the largest rate among the five receivers

pilot spacing, payload bits per frame, and the fallback policy differ among the selected configurations, so the figures compare complete receiver policies as well as N and pilot spacing

Fig. 8 uses $N = 1024$, CP 64, code rate $\frac{1}{4}$, outer/inner pilot spacing 8/8, and a configured pilot percentage of 25.000%. The other three figures state their own selected configurations the same way.

$N=1024$ · pilot 25.000% · spacing 8/8

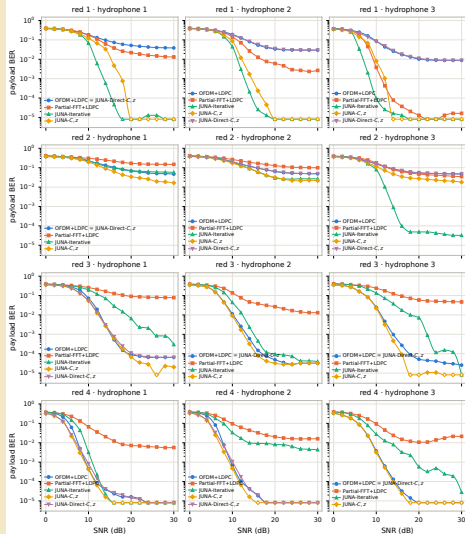


Fig. 8, $N = 1024$ of juna_joint_ieee.tex

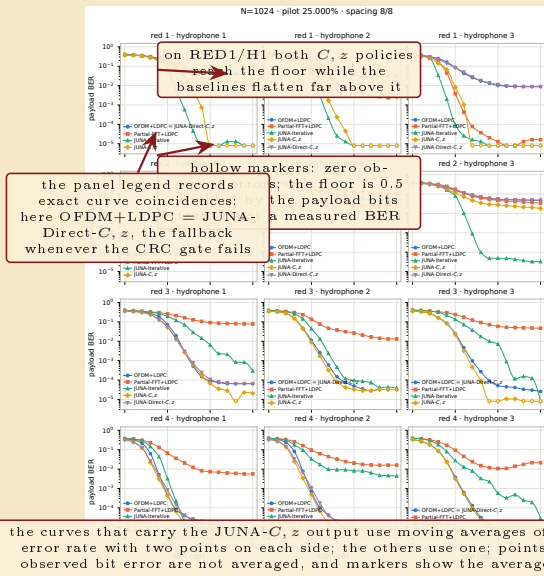


Fig. 8, $N = 1024$ of juna_joint_ieee.tex

N=512 · pilot 40.000% · spacing 5/5

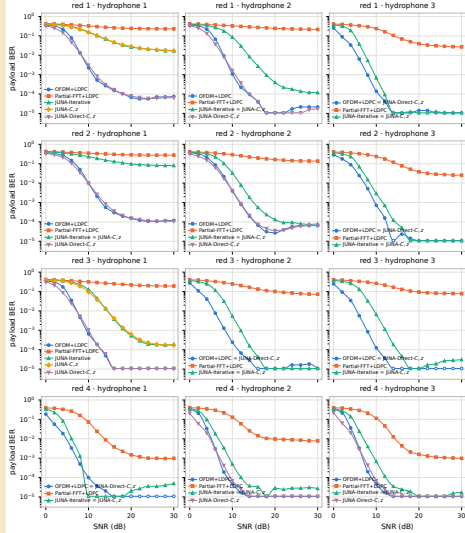
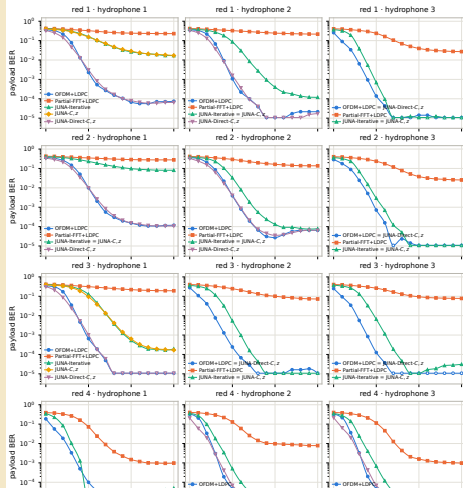


Fig. 9, $N = 512$ of juna_joint_ieee.tex

$N=512 \cdot \text{pilot } 40.000\% \cdot \text{spacing } 5/5$



$N = 512$, spacing 5/5, configured pilot percentage 40.000%. Here a JUNA- C, z candidate that fails the 16-bit CRC is replaced by the receiver's own Partial-FFT output; JUNA-Direct- C, z falls back to OFDM+LDPC.

Fig. 9, $N = 512$ of juna_joint_ieee.tex

N=1536 · pilot 34.286% · spacing 5/7

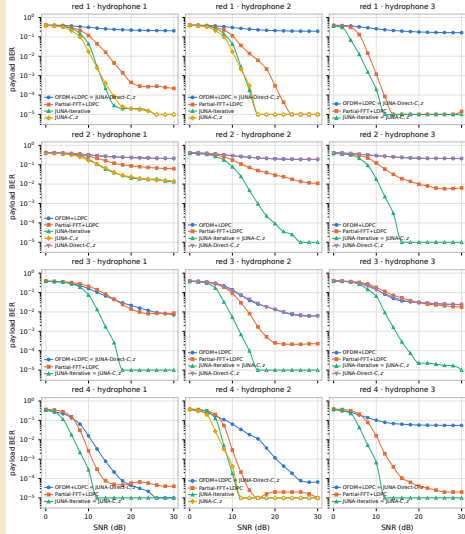
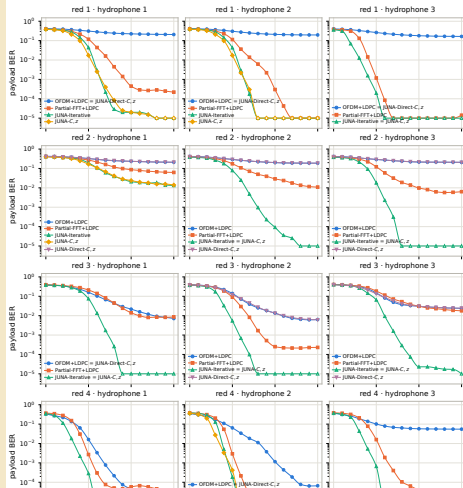


Fig. 10, $N = 1536$ of juna_joint_ieee.tex

$N=1536$ · pilot 34.286% · spacing 5/7



$N = 1536$, spacing 5/7, configured pilot percentage 34.286%. The fallback policies match the $N = 512$ comparison, so the four figures compare complete receiver policies as well as N and pilot spacing.

Fig. 10, $N = 1536$ of juna_joint_ieee.tex

N=2048 · pilot 30.000% · spacing 5/10

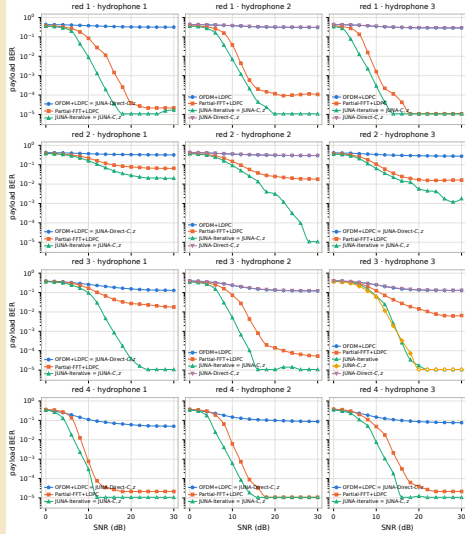
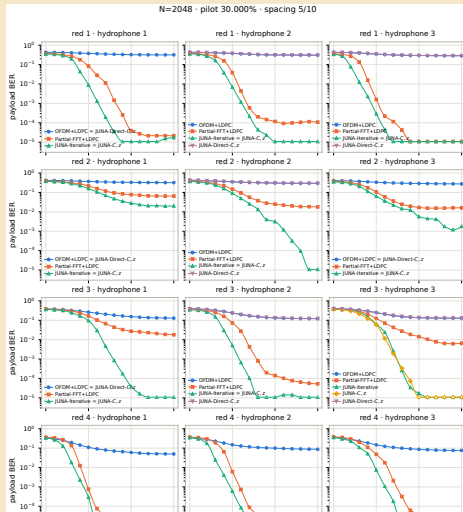


Fig. 11, $N = 2048$ of juna_joint_ieee.tex



$N = 2048$, spacing 5/10, configured pilot percentage 30.000%.
The same reading applies: hollow markers are the measurement floor, and the legends record exact curve coincidences.

Fig. 11, $N = 2048$ of juna_joint_ieee.tex

The quantity on the vertical axis

$$R_{1\text{ s}} = \frac{n_{\text{ok}} B_{\text{payload}}}{32 (1\text{ s})} \quad (62)$$

n_{ok} : the number of frames
with zero payload bit errors,
out of 32 transmitted

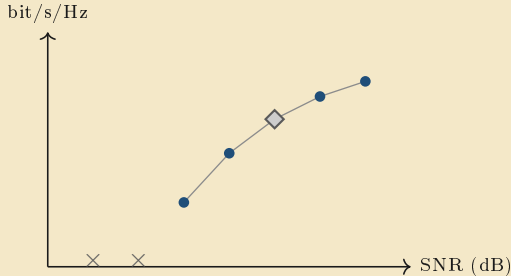
B_{payload} : the payload
bits per frame of the
selected configuration

the common one-second
denominator: every
configuration is scored
per second, not per frame

the vertical axis gives this effective payload rate di-
vided by the 9.6 kHz bandwidth, in bit/s/Hz

A frame counts only when every payload bit is exact. Hence the curve rewards configurations that both carry many bits and decode them without error.

One panel



only sampled SNR values are shown;
gray segments between adjacent
positive-rate points guide the eye

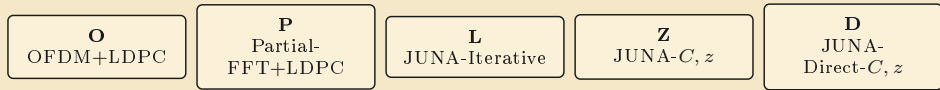
a gray cross on the rate axis marks
no recovered frame at that SNR

each point is the best ob-
served rate among the eigh-
teen admitted configurations

*The panel here is schematic; the printed figure carries the data. Every point answers one question:
at this path and SNR, what is the largest zero-error payload rate any admitted configuration and
receiver achieved?*

Point color and the receiver letters

point color identifies the receiver when one receiver alone gives the maximum



the same five letters name the receivers in both captions and in the Fig. 7 maps

When no single receiver wins, the point turns gray and its shape says who tied — the next slide.

The tie markers

a gray marker marks an exact receiver tie, and its shape gives the tied receivers



diamond

L=Z



square

O=D



triangle

O=Z=D



star

O=L=Z=D

an exact tie means equal best observed rate, not similar rate;

Fig. 7 prints the tied receivers in the cell

A tie is common where the frame decodes cleanly for several receivers at once; the tie says the receivers agree, not that the comparison failed.

The ribbon



the ribbon below the panel

the ribbon gives the complete selected configuration at each SNR

its key lists N , outer/inner spacing, and actual pilot percentage

a light-gray ribbon cell marks no recovered frame

The ribbon is where configuration changes show: when the best point switches from one N or pilot spacing to another, the ribbon cell changes with it.

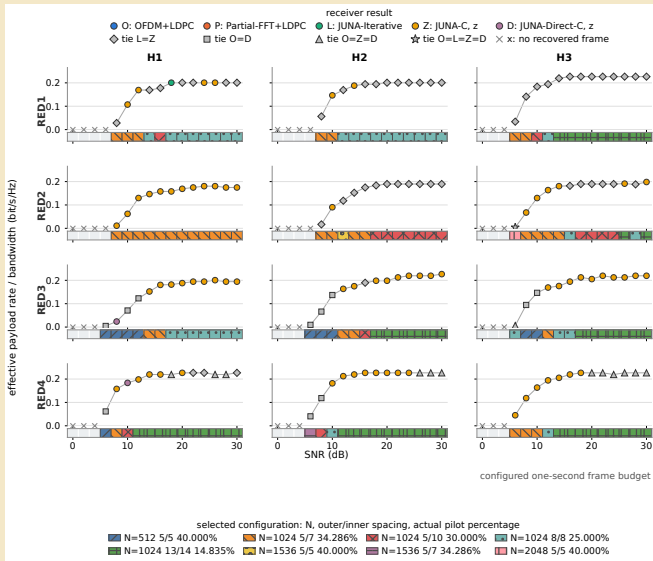


Fig. 6 of juna_joint_ieee.tex

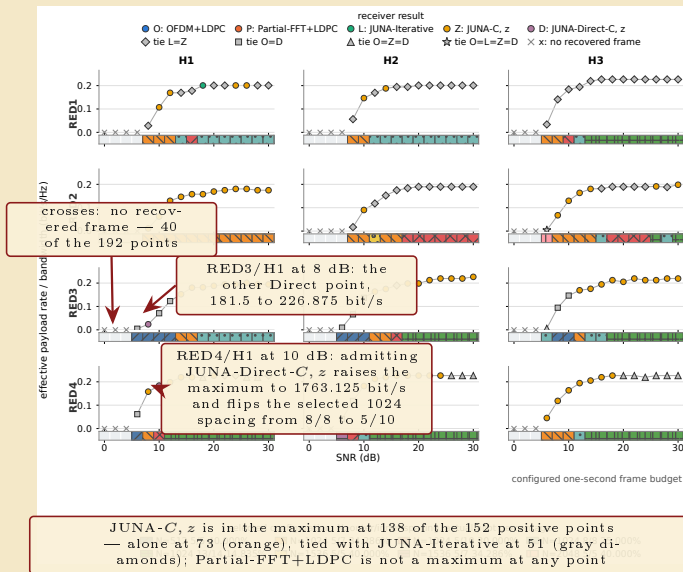


Fig. 6 of juna_joint_iees.tex

Figure 7: the two maps

upper map

Z L=Z 0=D × D

the upper map prints the exact receiver result in every cell; equality signs show exact ties

lower map

1024 8/8 512 10/10 2048 40/40

the lower map gives the complete selected configuration

in either map, a cross marks no recovered frame

The reading loop: find a point in Fig. 6, then read the same path and SNR cell in Fig. 7 — the upper map names the receivers, the lower map names the configuration. The maps here are schematic; the printed figure carries the data.

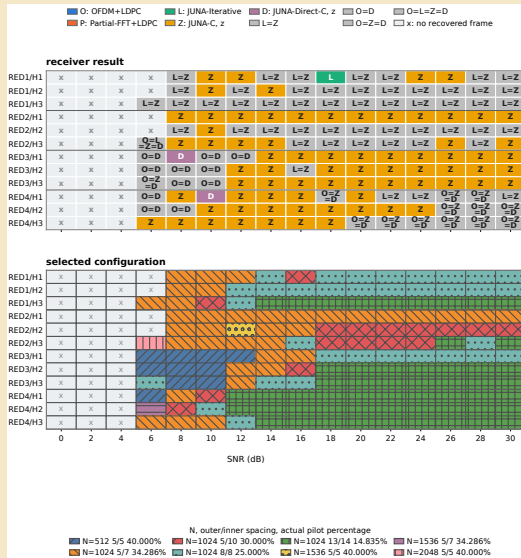


Fig. 7 of juna_joint_ieee.tex

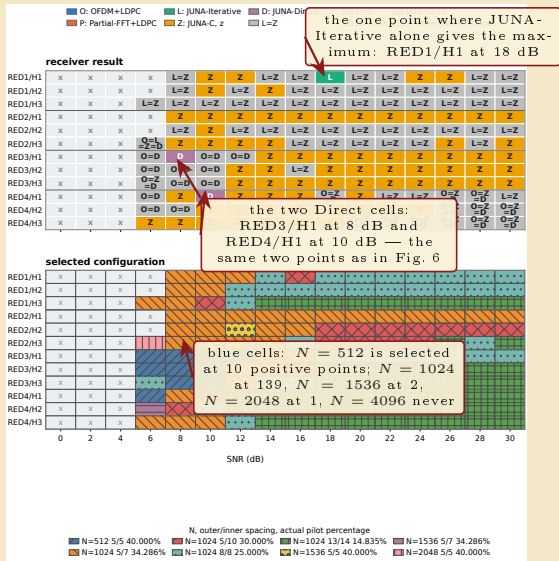


Fig. 7 of juna_joint_ieee.tex

Reading the Red result figures

the twelve panels are three hydrophones of one array observed in four segments of one deployment (Fig. 5)

the payload bit error rate figures show the same twelve paths at one selected configuration per N

every point: the best observed zero-error payload rate at one path and SNR (62)

color names the sole winner; gray shapes name exact ties

the ribbon and the lower map name the selected configuration

the upper map prints every receiver result, ties included

receiver and configuration should be compared together: no one choice gives every observed maximum